

Abhishek Panthalingal

Bellevue, WA | abhisheknr4@gmail.com | 765-631-9411 | linkedin.com/in/6foot4

SUMMARY

Lead Product Technologist with 9 years of experience architecting cloud-native platforms on Azure, Databricks, and Snowflake, paired with deep frontend expertise in React, Next.js, and TypeScript. Currently delivering executive analytics, supply chain, and hardware-integrated systems for T-Mobile US and TOMRA through Rhombuz LLC, including a multi-year T-Mobile relationship spanning network analytics, cloud platform engineering, and supply chain. Prior enterprise engagements with Albertsons and Comcast. Architecture ownership and technical leadership on client-facing engagements for over 5 years, across warehouse-scale analytics, cloud infrastructure, and hardware-integrated field applications.

TECHNICAL SKILLS

Cloud and Data Platforms: Microsoft Azure (App Service, Container Apps, Functions, Cosmos DB, PostgreSQL, Application Insights, Entra ID, Private Endpoints), Databricks, Snowflake, AWS, Docker, Kubernetes, GitLab CI/CD, KQL, SQL, Common Table Expressions (CTEs)

AI and Machine Learning: LLM application integration, anomaly detection, prompt engineering, prompt-injection guardrails, MCP-based tooling, AI-assisted development

Frontend Development: React.js, Next.js, React Native, Redux, TypeScript, JavaScript, ES6, HTML5, CSS3, Webpack, Material UI, Responsive Design, Accessibility (a11y), Web Performance Optimization

Backend and APIs: Node.js, Express.js, GraphQL, REST APIs, MongoDB, Data Visualization

Mobile and Device Integration: iOS, iPadOS, Android, Bluetooth Low Energy (BLE), Industrial Hardware Integration

Practices: Software Architecture, Technical Leadership, Agile, Scrum, Test-Driven Development (TDD), Code Review, Mentoring, UI/UX Design, Web Security

PROFESSIONAL EXPERIENCE

Rhombuz LLC | Lead Product Technologist

Seattle, WA | April 2025 to Present

Technical leadership across client engagements, with architecture ownership, direct client accountability, and responsibility for technical hiring. Engaged as a contractor to T-Mobile US and TOMRA.

Client: T-Mobile US — Network Analytics for Executive Decision-Making

- Architected an executive analytics platform used by directors, VPs, and SVPs to inform network build, performance, and planning decisions across the carrier's national footprint
- Designed the data model and URL-driven state architecture around a five-level geographic hierarchy (National, SuperRegion, Region, Area, Market), supporting 80+ markets within a single application
- Built aggregate metric views over Snowflake and Azure PostgreSQL ingesting millions of rows per day, using conditional aggregation and CTE-based reporting patterns
- Instrumented the platform end to end in Azure Application Insights and authored KQL analytics covering daily and weekly active users, cross-week retention cohorts, and dwell time by dashboard area
- Identified development traffic contaminating production engagement metrics and corrected the reporting baseline used for adoption and roadmap decisions
- Implemented automated export and distribution of reporting in editable Microsoft Word and PowerPoint formats for downstream recipients

Client: T-Mobile US — 5G Network Anomaly Detection

- Built a market-level anomaly detection pipeline in Databricks against Azure PostgreSQL, applying statistical scoring to flag underperforming markets and month-over-month 5G traffic decline
- Integrated a Databricks-hosted Llama 70B model to generate plain-English root-cause analysis and a recommended next action for each flagged market, making the output actionable for non-analyst network teams
- Established credential handling through Databricks secret scopes and JDBC-based ingestion from Azure PostgreSQL

Client: T-Mobile US — Cloud Platform Engineering

- Led migration of a production service from Azure Function App to App Service, including private endpoint and private DNS cutover across non-production and production environments
- Diagnosed and resolved authentication failures spanning Azure Container Apps, Cosmos DB, and Entra ID Easy Auth configuration
- Managed GitLab CI/CD pipelines, Azure service principals, and container registry credentials across non-production and production environments

Client: T-Mobile US — Supply Chain Control Tower

- Built control tower systems providing end-to-end visibility across the supply chain for device distribution and servicing, covering the sales and upkeep of iPhone devices across the carrier's retail footprint

- Own the user interface and API layer end to end, holding full architectural responsibility for both tiers from contract design through implementation and delivery
- Designed the API contracts exposing device and inventory state from upstream supply chain systems to the operator-facing application

Client: TOMRA — Recycling Collection Infrastructure

- Designed and delivered an iPadOS application supporting the commissioning and daily operation of recycling collection points, enabling TOMRA to bring deposit-return service to client locations that were not previously served
- Integrated the application with field hardware, including industrial weighing equipment (OHAUS Defender 5000) and operator display units (Optima OP-910) over Bluetooth Low Energy
- Enabled deployment to 5 or more collection locations across California, with expansion to additional states underway, supporting processing of over 1,000 pounds of recycled material daily
- Reduced the technical effort required to commission a new collection point, lowering the barrier to expanding recycling access
- Authored operator-facing setup and troubleshooting documentation used by staff at deployed locations

Internal: Engineering and Hiring

- Designed structured technical interview frameworks for full-stack and software engineering hires, including coding exercises, system design prompts, and behavioral rubrics
- Built an automated briefing pipeline processing overnight email and chat traffic, with prompt-injection guardrails
- Led exploratory architecture work on on-premises AI inference for customers with data-residency constraints

UST | Lead II, Software Engineering

Seattle, WA | January 2023 to March 2025

Technical leadership across enterprise client engagements, guiding development teams and managing client relationships.

Client: Albertsons — Frontend Development Lead

- Led onsite frontend development for a large-scale enterprise retail product
- Designed a scalable frontend architecture using React.js and Next.js, improving page load speed by 30 percent and reducing maintenance overhead by 25 percent
- Implemented Redux state management and applied AI-assisted tooling to increase test coverage by over 20 percent against organizational targets
- Led UX design implementation and web security hardening, and managed cross-functional team dynamics and direct client communication

Client: Comcast — Lead Software Engineer

- Led end-to-end development of a minimum viable product across multiple interdisciplinary teams in an Agile and TDD environment
- Implemented Docker and Kubernetes containerization and built CI/CD pipelines, reducing deployment times by 40 percent
- Conducted code review and technical mentoring

Client: T-Mobile US — Lead Software Engineer

- Designed and developed an executive dashboard supporting high-level strategic decision-making
- Refactored and optimized a complex data visualization library, raising Lighthouse performance scores from 75 to 95 or above
- Received cross-departmental recognition from managers on multiple teams for the dashboard's design and technical execution

Piktorlabs | Senior Design Technologist / Product Engineer

Seattle, WA and Bangalore, India | June 2019 to December 2022

Lead engineer on multiple engagements, including contract work for T-Mobile US and end-to-end ownership of the GoTag product.

- Led GoTag from design through release, including compliance and release testing for publication to the ADP Marketplace
- Built GoTag as a cross-platform React Native application for Android and iOS, reducing development effort and redundancy by 40 percent
- Implemented Redux-based state management to improve performance and reduce data dependency
- Improved application performance by 20 percent through network request optimization, and deployed applications using Docker and Kubernetes for scalable delivery

Moonraft Innovation Labs | Software Engineer

Bangalore, India | September 2017 to May 2019

- Developed features for NorthStar, an application built for UST serving a user base of over 17,000
- Delivered the organization's first React.js project, driving internal adoption of the library; ran workshops and mentored junior developers

EDUCATION

UVCE, Bangalore University | Bachelor of Engineering, Mechanical Engineering | 2017 | Graduated with Distinction

DesignBoat UI/UX School | Bootcamp in UI/UX Design, Bangalore, India | 2021